Abstract:

Hand paralysis is a common and life-altering complication experienced by stroke patients, often limiting their ability to perform daily activities independently. To address this issue, this project presents a vision-based hand exercise recognition and feedback system that supports rehabilitation without relying on wearable sensors. The proposed system uses a standard camera to capture hand movements and applies computer vision techniques for real-time hand landmark extraction. Using MediaPipe-based hand landmark detection, spatial features are extracted and classified using machine learning algorithms to recognize multiple predefined hand rehabilitation exercises, including wrist stretching, finger flexion, grip strengthening, thumb movements, and abduction–adduction motions. The system achieves high recognition accuracy while maintaining real-time performance. A hardware-based feedback module is integrated to provide real-time visual, auditory, and wireless feedback to the user, enhancing interaction and engagement during rehabilitation. The proposed system is non-invasive, and suitable for home-based rehabilitation environments. Experimental results demonstrate the effectiveness and reliability of the system for real-time hand exercise monitoring.